Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building secure SaaS workflows, backend automation, and production systems that reduce operational friction across AI, healthcare-adjacent, and business process platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Architected a **multi-tenant AI education platform** on **Azure SQL** and Blob Storage with row-level security for **500+ users**, reducing dashboard latency by **71%** (**1.2s to 350ms**) through schema and indexing optimization.
- Built an async data integration and LLM job orchestration pipeline using **FastAPI, Supabase, SQL Server, and cloud storage**, processing **5K+ monthly jobs** with **3x** higher throughput and **99%+** production stability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Engineered an **AI-powered grading platform** with **GPT-4o, Claude 3.7, Qwen-VL**, and OCR to evaluate text and handwritten submissions for **150+ instructors**, handling **10K+ grading requests/day**.
- Designed a concurrent batch-processing pipeline for **PDF, DOCX, and ZIP** ingestion using **Flask, MySQL, AWS RDS, and REST APIs**, cutting feedback turnaround time by **60%** and supporting **1K+** files per batch run.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built cross-platform data-sharing software with a **microservices architecture**, automating VM builds and **20+** unit/integration test suites using **Jenkins, PyTest, and Mockito** across Windows, Linux, and macOS environments.
- Optimized API and ETL workflows with **AWS Glue, S3, and OOP patterns**, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed a **MERN-based resource management platform** with **WebSockets**, REST APIs, and microservices across **8** business workflows, synchronizing **2M+ records** and reducing review turnaround by **25%**.
- Automated **CI/CD** with **Jenkins** and Docker and integrated **Azure SQL** plus analytics instrumentation, improving release reliability to **95%+** and reducing resource wastage by **42%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **Flask/Firebase** booking and payment workflows for a cross-platform service app handling **50K+ records**, improving confirmation speed by **40%** through real-time sync and cutting manual refund handling by **70%** with webhook automation.
- Integrated multi-currency payment, coupon, and analytics workflows using **Databricks, Spark, and webhooks**, supporting sub-**200ms** booking APIs and hourly reporting for pricing, staffing, and operational decisions.

Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++
Backend, APIs & Workflows: Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, Microservices
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS (RDS, S3, EC2), Microsoft Azure, Firebase
Testing, CI/CD & Ops: PyTest, Mockito, Jenkins, GitHub Actions, Docker, CI/CD, Monitoring, Debugging
Engineering: Distributed Systems, ETL Pipelines, Performance Optimization, Workflow Automation, Secure SaaS Platforms

Projects

HMIS – Hospital Management Information System Live | *ABDM APIs, Appwrite, Twilio, Sentry, TypeScript*

- Built a healthcare workflow platform integrating **ABDM APIs** to support **100+ daily appointments**, reduce patient record retrieval by **2x**, and cut administrative workload by **30%** through consent-aware records and automated notifications.

Chat2SQL | *SQL, PostgreSQL, Flask*

- Designed a schema-aware SQL generation workflow that translated natural-language questions into optimized queries with **82% accuracy**, improving validation speed and analyst-style reporting on relational datasets.

FinNexus Live | *Next.js, Kubernetes, MongoDB, Plaid*

- Engineered a secure fintech SaaS platform for **50K+ users** and **1M+ transactions**, adding real-time account updates, CI/CD-driven release workflows, and tokenized integrations for bank linking and transaction visibility.